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# SWEN90010 - High Integrity Systems Engineering

Safety Engineering, HAZOP, Fault Tree Analysis

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# **SAFETY ENGINEERING**

# Safety

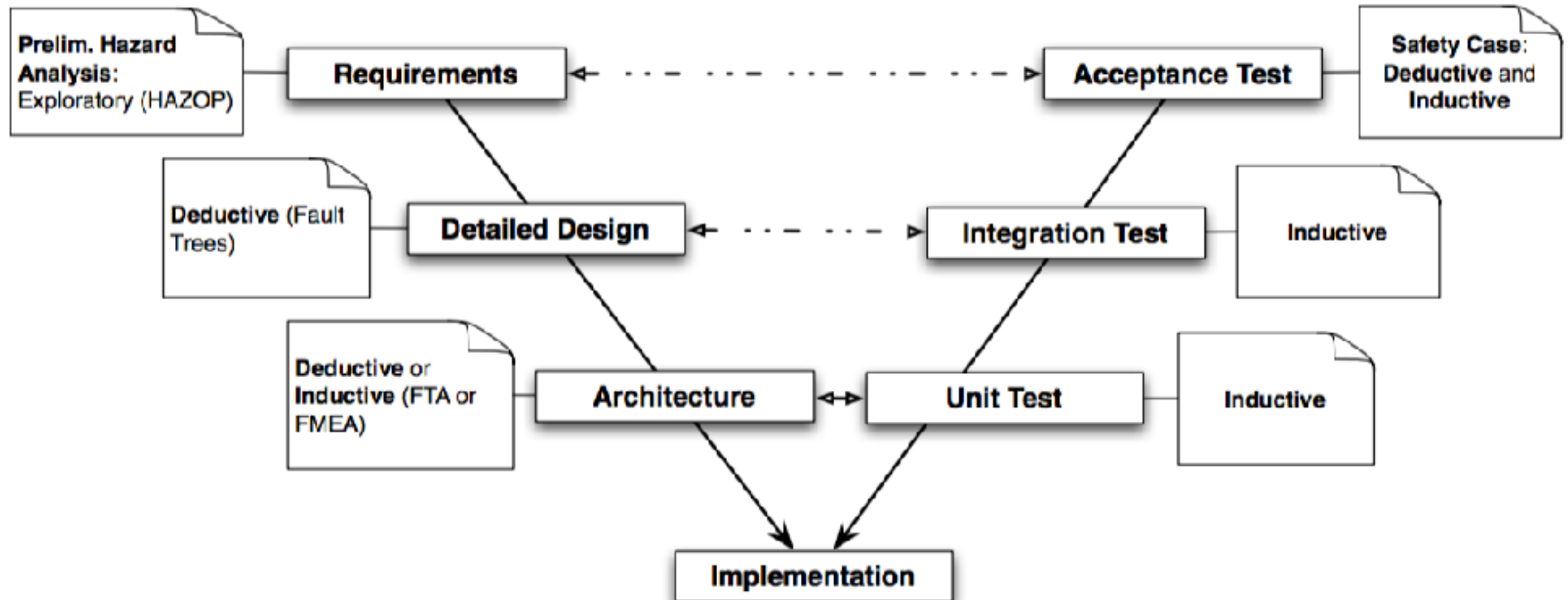
In light of the above, a more precise definition:

**software and hardware used under correct  
operating conditions don't cause unacceptable  
harm to people or environment.**

# Safety Engineering Process



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# Safety Engineering



How do we engineer safe systems?

## History

Safety engineers are **experts** in their domain:  
medical, rail signalling, aviation etc.

Safety engineers are experts in past **accidents**,  
**incidents** and **failures**.

It is difficult to guard against what you can't predict.

Is why air crashes are so thoroughly investigated.

# Hazards

Safety engineers examine past accidents etc.

looking for **hazards**:

things that could lead to (future) accidents

Then design the system to be safe in the face of these things, reducing the chance of accidents happening.

Tends to be the same factors over and over again.

Doing it properly requires determining **causation**.

Not enough to know that the accident occurred because a device failed; need to understand **why** it failed, what **caused** it to fail and e.g. why the pilot didn't notice.

# Causality

Not correlation.

Often contested, not always clear cut.

Determined by **counterfactual** reasoning.

**A counterfactual:** “if *A* hadn’t happened, what would be the case ...”

“*A* is a cause of *B* when, **if *A* hadn’t happened, *B* wouldn’t have happened.**”

# Causality is not Correlation



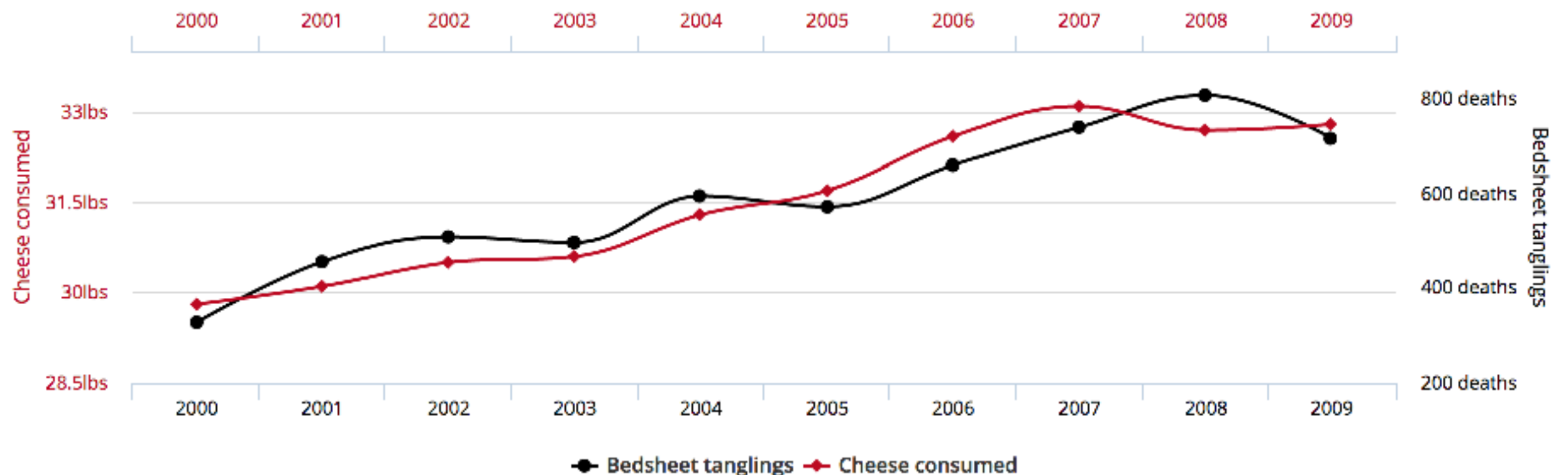
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<https://www.tylervigen.com/spurious-correlations>

**Per capita cheese consumption**  
correlates with

**Number of people who died by becoming tangled in their bedsheets**

Correlation: 94.71% ( $r=0.947091$ )



tylervigen.com

Data sources: U.S. Department of Agriculture and Centers for Disease Control & Prevention

Absence of causation is revealed by  
counterfactual reasoning.

# Causality is Not Correlation



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**Confounding:** deaths by drowning correlated  
with ice-cream consumption

(common cause: warm weather)

**Reversed Causality:** barometer reading always  
drops before a storm

(but the barometer reading doesn't cause the storm)

Absence of causation is revealed by  
counterfactual reasoning.

# Past Causes as Hazards

Safety engineers study past accident causes  
to determine hazards for new systems

Works because new systems tend to be  
similar to old ones.

e.g. entirely new modes of transport  
tend to be invented rarely.

But must be done **systematically**  
by following a process, to ensure **repeatability**.

(This is true of all engineering activities, of course.)

# Safety Engineering Tasks



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## **Preliminary Hazard Analysis**

Based on requirements and early design:  
“what could go wrong?”

Hazard brainstorming.

e.g. HAZOP

e.g. Fault-Tree  
Analysis

## **Design**

Given those hazards, design the system  
to eliminate or mitigate them  
(depending on their risk)

e.g. Formal  
Specification

e.g. Program  
Verification

## **Implementation**

Was it done right?



# HAZOP: HAZARDS AND OPERABILITY STUDY

# HAZOP Study

A method for Preliminary Hazard Analysis

Exploratory analysis, **systematically** brainstorming  
what could go wrong

Introduced originally to understand hazards in  
chemical processes, 1950s

Adopted in safety-critical software engineering  
due to its effectiveness

# HAZOP Overview

**Input:** high-level system description as  
a collection of **design items**

**Design item:** an intended behaviour, e.g.  
an event in a process or a state transition

“when sensor X is triggered, valve B is  
closed immediately”

Each design item is systematically  
**mutated** to analyse what could happen  
when the intended behaviour doesn't occur,  
and to determine the **risk**  
(consequences and frequency).

# Assessing Hazard Risks

## IEC - 61508

| Frequency  | Consequence  |          |          |            |
|------------|--------------|----------|----------|------------|
|            | Catastrophic | Critical | Marginal | Negligible |
| Frequent   | I            | I        | I        | II         |
| Probable   | I            | I        | II       | III        |
| Occasional | I            | II       | III      | III        |
| Remote     | II           | III      | III      | IV         |
| Incredible | IV           | IV       | IV       | IV         |

**Class I** Intolerable risk.

**Class II** Undesirable risk and tolerable only if risk reduction is impractical.

**Class III** Tolerable risk if the cost of risk reduction would exceed the improvement gained.

**Class IV** Negligible risk.

Safety engineering about moving from **Class I** towards **IV**

# Risk: Consequence x Freq.



| Frequency  | Consequence  |          |          |            |
|------------|--------------|----------|----------|------------|
|            | Catastrophic | Critical | Marginal | Negligible |
| Frequent   | I            | I        | I        | II         |
| Probable   | I            | I        | II       | III        |
| Occasional | I            | II       | III      | III        |
| Remote     | II           | III      | III      | IV         |
| Incredible | IV           | IV       | IV       | IV         |

# Consequences and Freq.

## Frequency

| Category   | Definition                         | Range (failures per year) |
|------------|------------------------------------|---------------------------|
| Frequent   | Many times in lifetime             | $> 10^{-3}$               |
| Probable   | Several times in lifetime          | $10^{-3}$ to $10^{-4}$    |
| Occasional | Once in lifetime                   | $10^{-4}$ to $10^{-5}$    |
| Remote     | Unlikely in lifetime               | $10^{-5}$ to $10^{-6}$    |
| Improbable | Very unlikely to occur             | $10^{-6}$ to $10^{-7}$    |
| Incredible | Cannot believe that it could occur | $< 10^{-7}$               |

## Consequences

| Category     | Definition                    |
|--------------|-------------------------------|
| Catastrophic | Multiple loss of life         |
| Critical     | Loss of a single life         |
| Marginal     | Major injuries to one or more |
| Negligible   | Minor injuries at worst       |

# Assessing Hazard Risks

## IEC - 61508

| Frequency  | Consequence  |          |          |            |
|------------|--------------|----------|----------|------------|
|            | Catastrophic | Critical | Marginal | Negligible |
| Frequent   | I            | I        | I        | II         |
| Probable   | I            | I        | II       | III        |
| Occasional | I            | II       | III      | III        |
| Remote     | II           | III      | III      | IV         |
| Incredible | IV           | IV       | IV       | IV         |

**Class I** Intolerable risk.

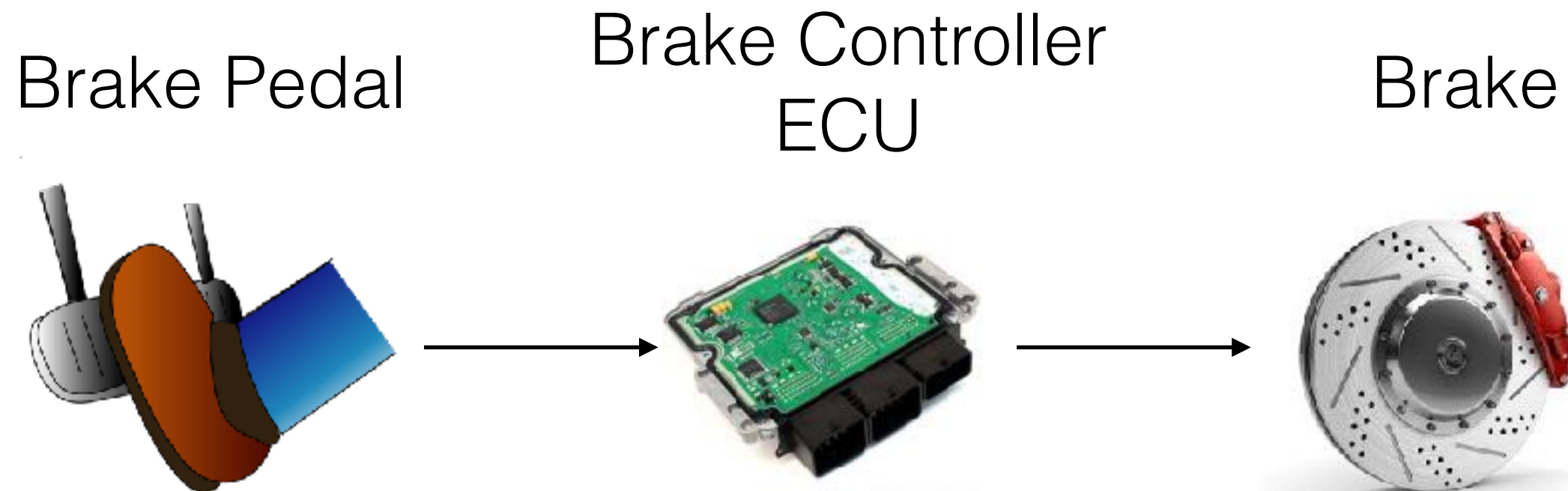
**Class II** Undesirable risk and tolerable only if risk reduction is impractical.

**Class III** Tolerable risk if the cost of risk reduction would exceed the improvement gained.

**Class IV** Negligible risk.

Safety engineering about moving from **Class I** towards **IV**

# Example Scenario



# Design Items

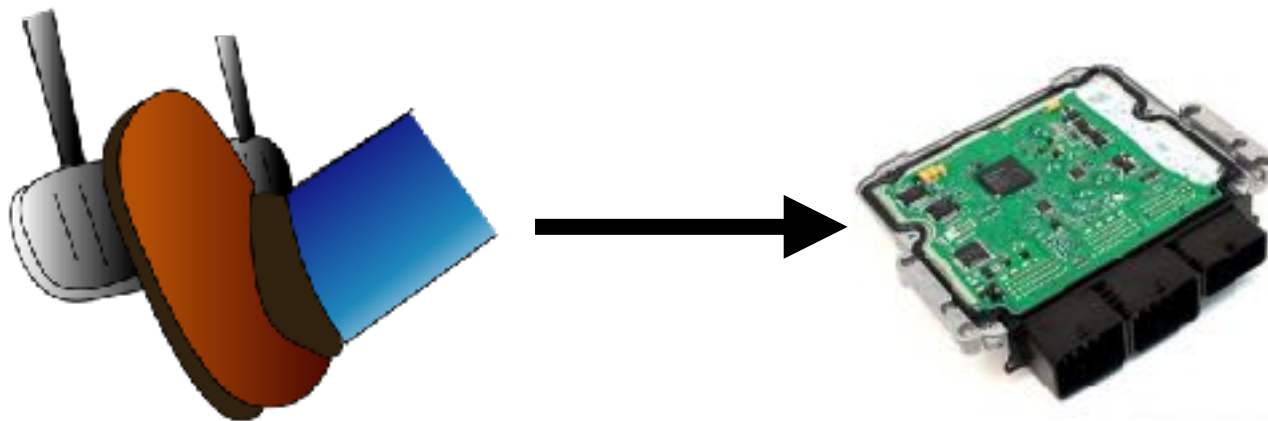
Depends on level of abstraction.

“when detecting a rapid increase in brake pedal pressure, the ECU will apply the brake with full force to implement emergency braking.”

ECU

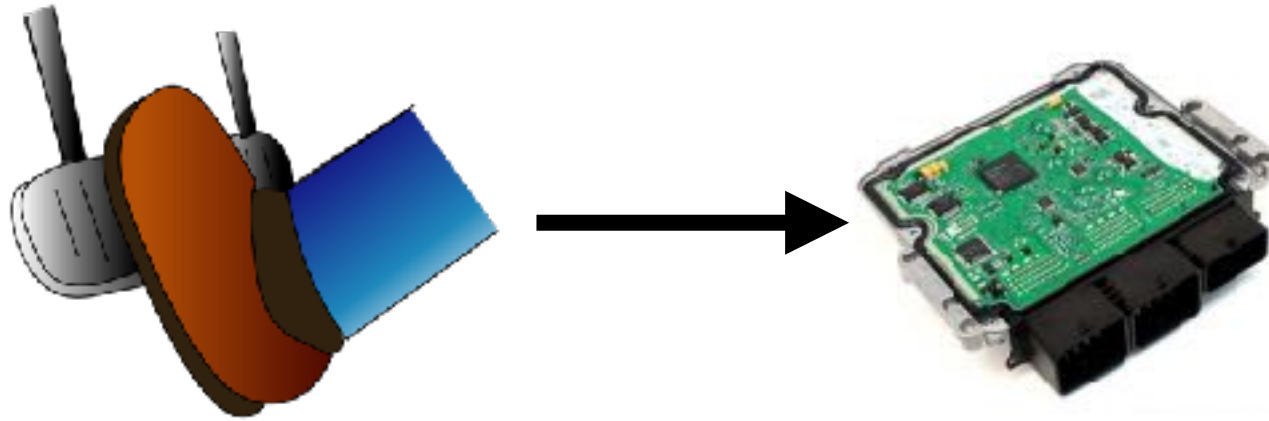


“the Brake Controller control loop is scheduled to run once every 1/10th of a second”



“the brake controller ECU receives a signal from the brake pedal when its foot pressure changes”

# What-Ifs to Consider



## Design Item

“the brake controller ECU receives a signal from the brake pedal when its foot pressure changes”

**What if** there is **no** signal from the brake pedal?

**What if** the signal comes **early**?

**What if** the signal comes **late**?

For each, think about the potential hazards that could arise and rate their **consequences** (catastrophic, critical, marginal, negligible).

# Hazards and Consequences



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**What if** there is **no** signal from the brake pedal?

1. car fails to stop. Consequence: Catastrophic
2. car gets stuck. Consequence: Critical (?)

**What if** the signal comes **early**?

1. nonsensical, not applicable (?)
2. car brakes too quickly (?)

**What if** the signal comes **late**?

1. car fails to stop in time. Consequence: Catastrophic
2. car fails to start in time. Consequence: Critical (?)

# HAZOP Guidewords



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Used to explore What-Ifs **systematically**

| Guide Word | Deviation                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| NO or NONE | This is the complete negation of the design intention. No part of the intention is achieved and nothing else happens. |
| MORE       | This is a quantitative increase.                                                                                      |
| LESS       | This is a quantitative decrease.                                                                                      |
| AS WELL AS | All the design intention is achieved together with additions.                                                         |
| PART OF    | Only some of the design intention is achieved.                                                                        |
| REVERSE    | The logical opposite of the intention is achieved.                                                                    |
| OTHER THAN | Complete substitution, where no part of the original intention is achieved but something quite different happens.     |
| EARLY      | Something happens earlier than expected relative to clock time.                                                       |
| LATE       | Something happens later than expected relative to clock time.                                                         |
| BEFORE     | Something happens before it is expected, relating to order or sequence.                                               |
| AFTER      | Something happens after it is expected, relating to order or sequence.                                                |

# HAZOP Process

for each **design item**

for each **guideword**

**interpret** the guideword against the design item  
(ask “what could it mean?”)

giving **deviations** from intended behaviour

determine potential **causes** of each deviation

potential **consequences**

**safeguards** in place to mitigate it

**risk**

design **recommendations**

# HAZOP Outcomes

Big spreadsheet of hazards

Lots of documentation

Useful for building **safety case**

And for analysing future safety incidents

(But somebody has to manually analyse it all)

Challenging, creative and interesting process

Iterate by feeding recommendations back into design

Since design recommendations will change  
**risk** ratings

## Overall Process

1. Identify design items
2. For each, do each guideword
  - causes
  - consequences
  - safeguards
  - risk
  - (design) recommendations



# FAULT TREE ANALYSIS

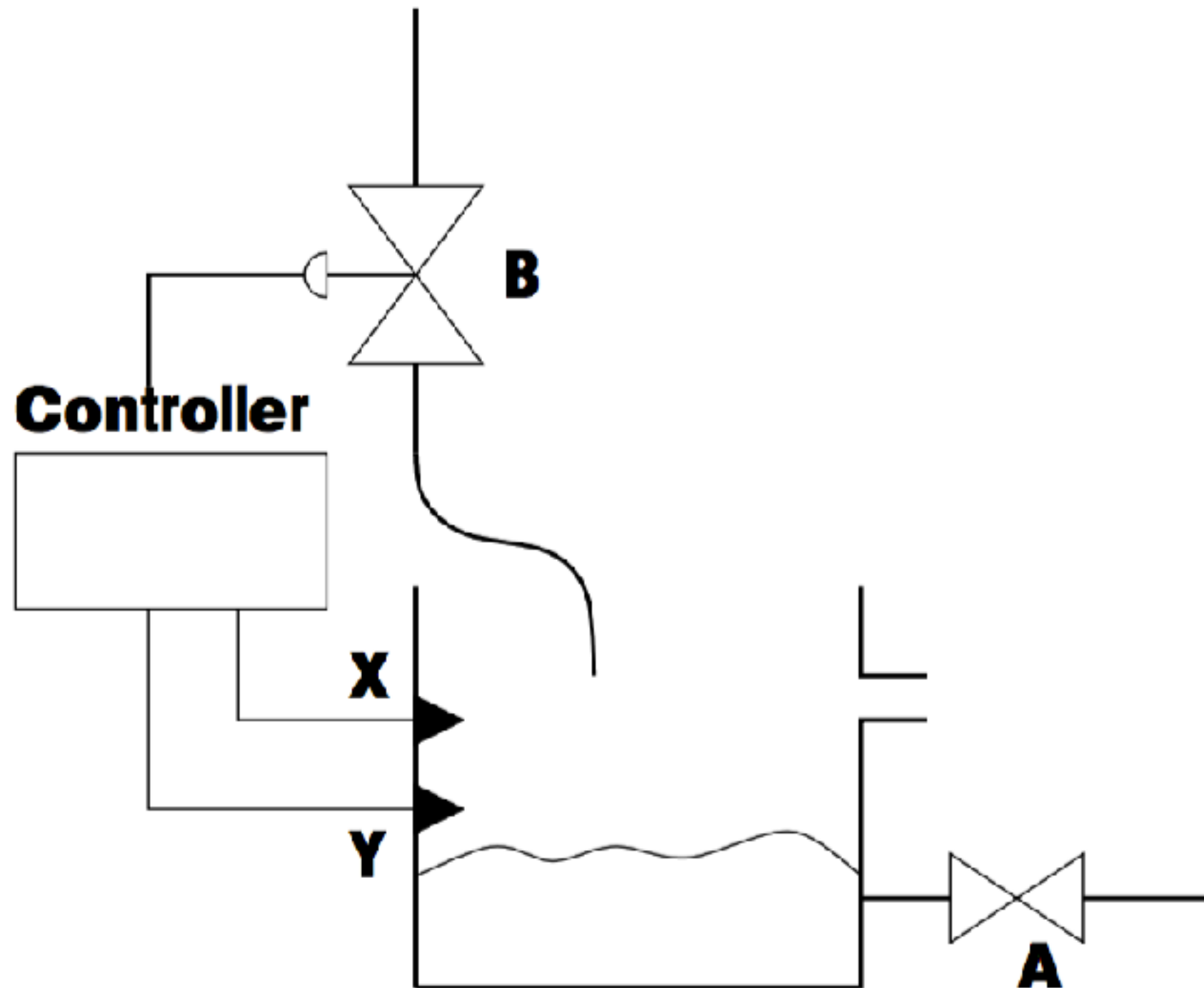
# Fault Tree Analysis

Does my system's design  
properly mitigate the hazards  
identified during PHA (e.g. HAZOP)?

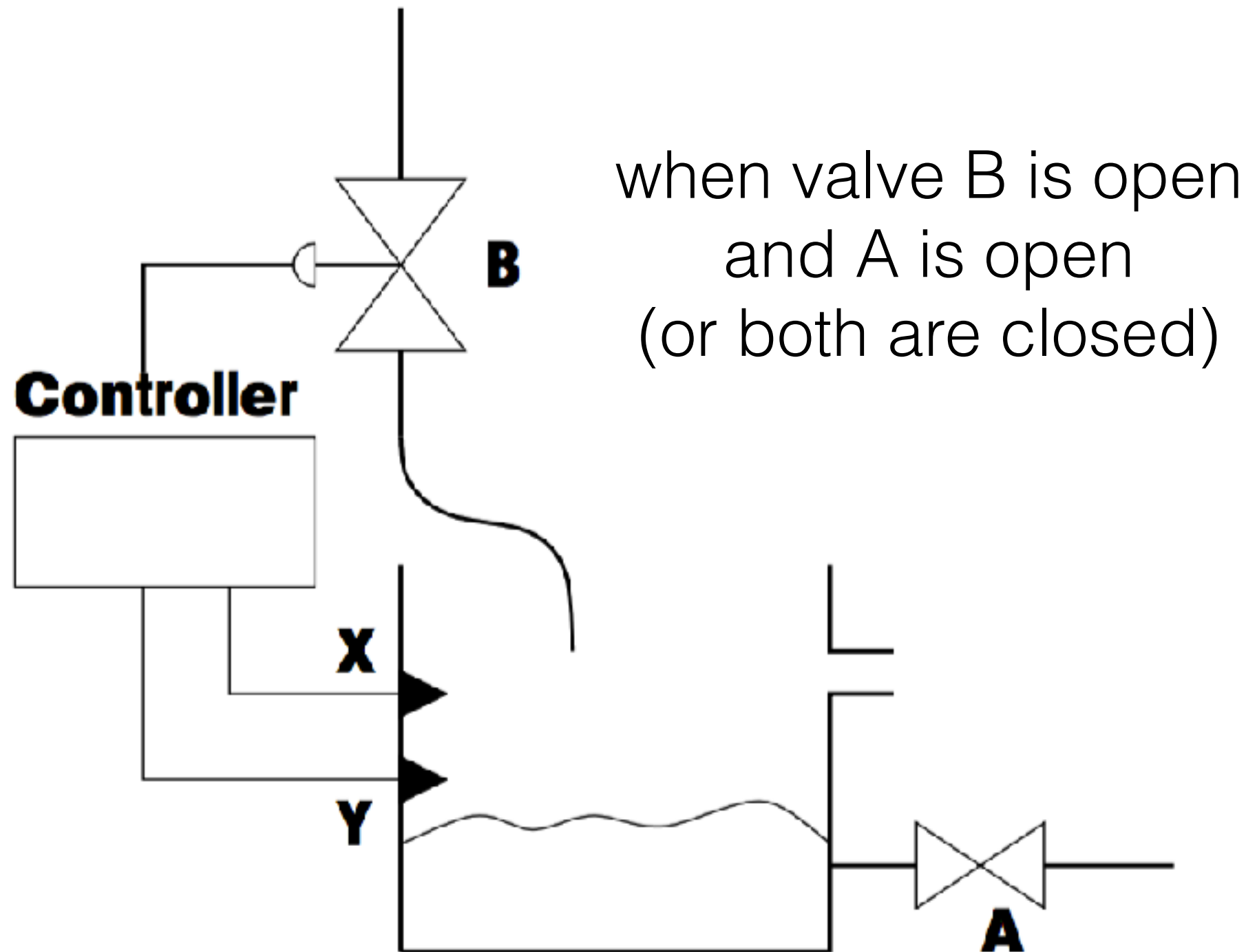
Analyses when and how hazards  
can happen, in the context  
of the system's design.

Is **deductive**, working **backwards**  
from hazards (or events that lead to them)  
to how they can be caused

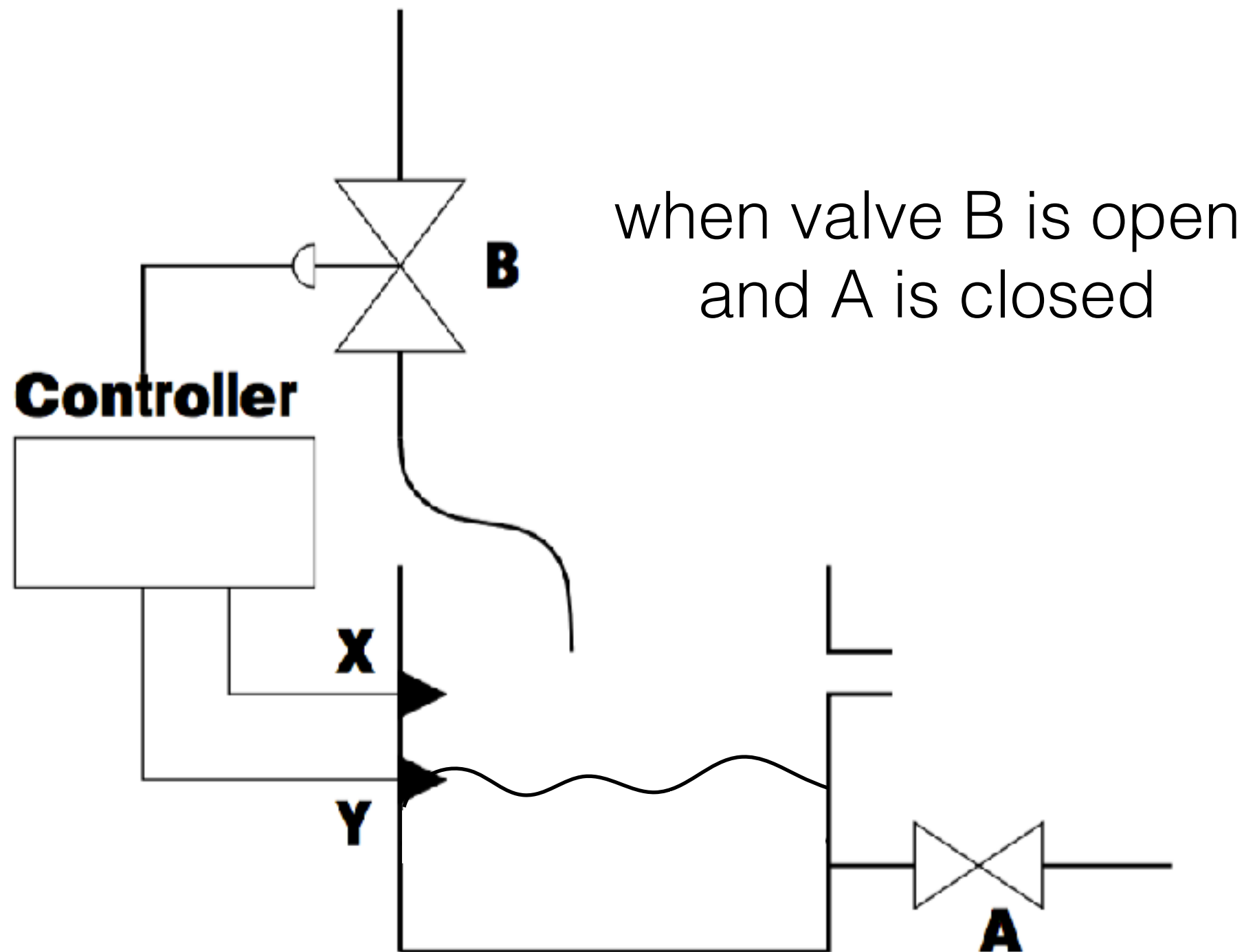
# Example: Chemical Mixing



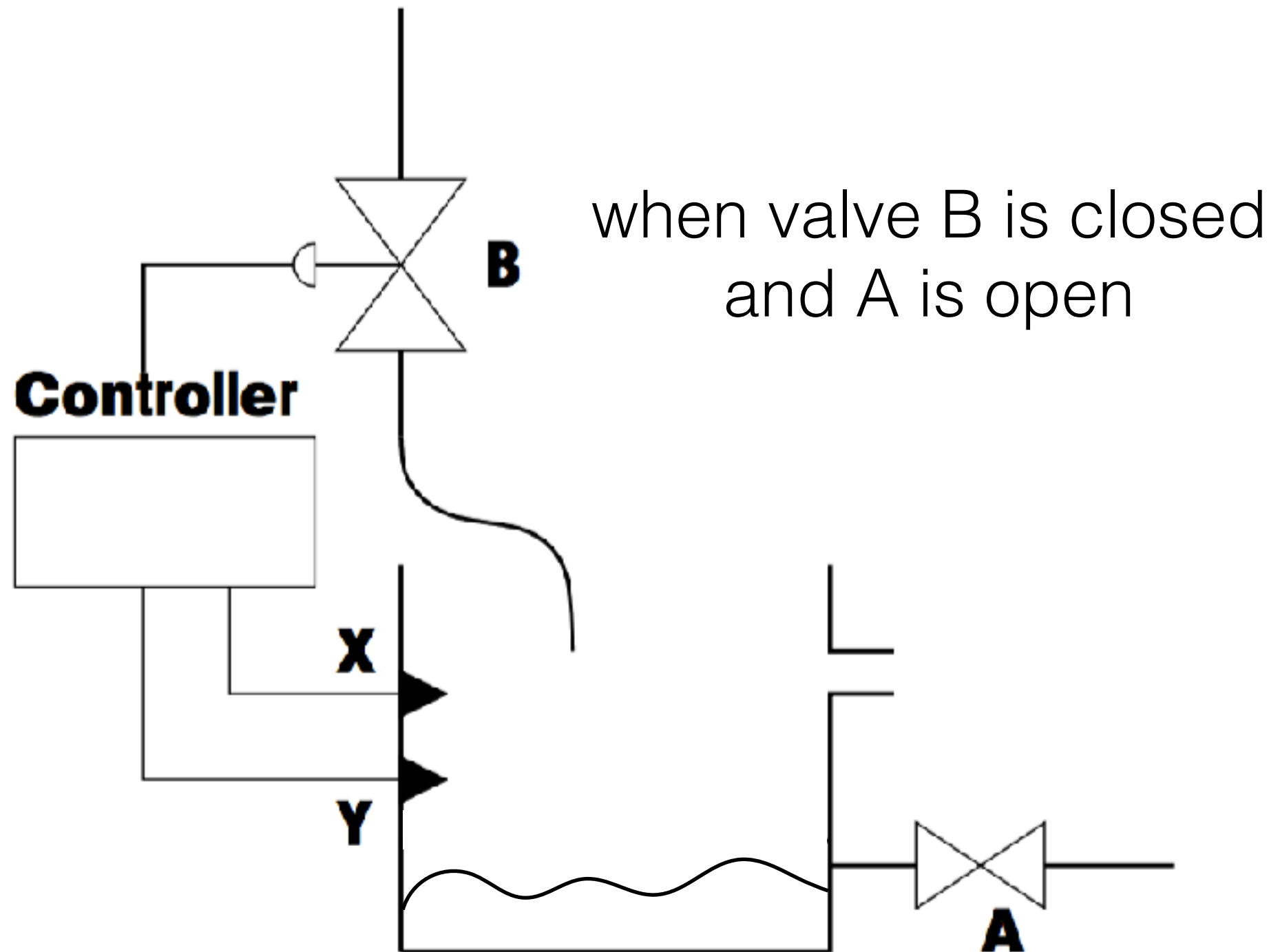
# Example: Chemical Mixing



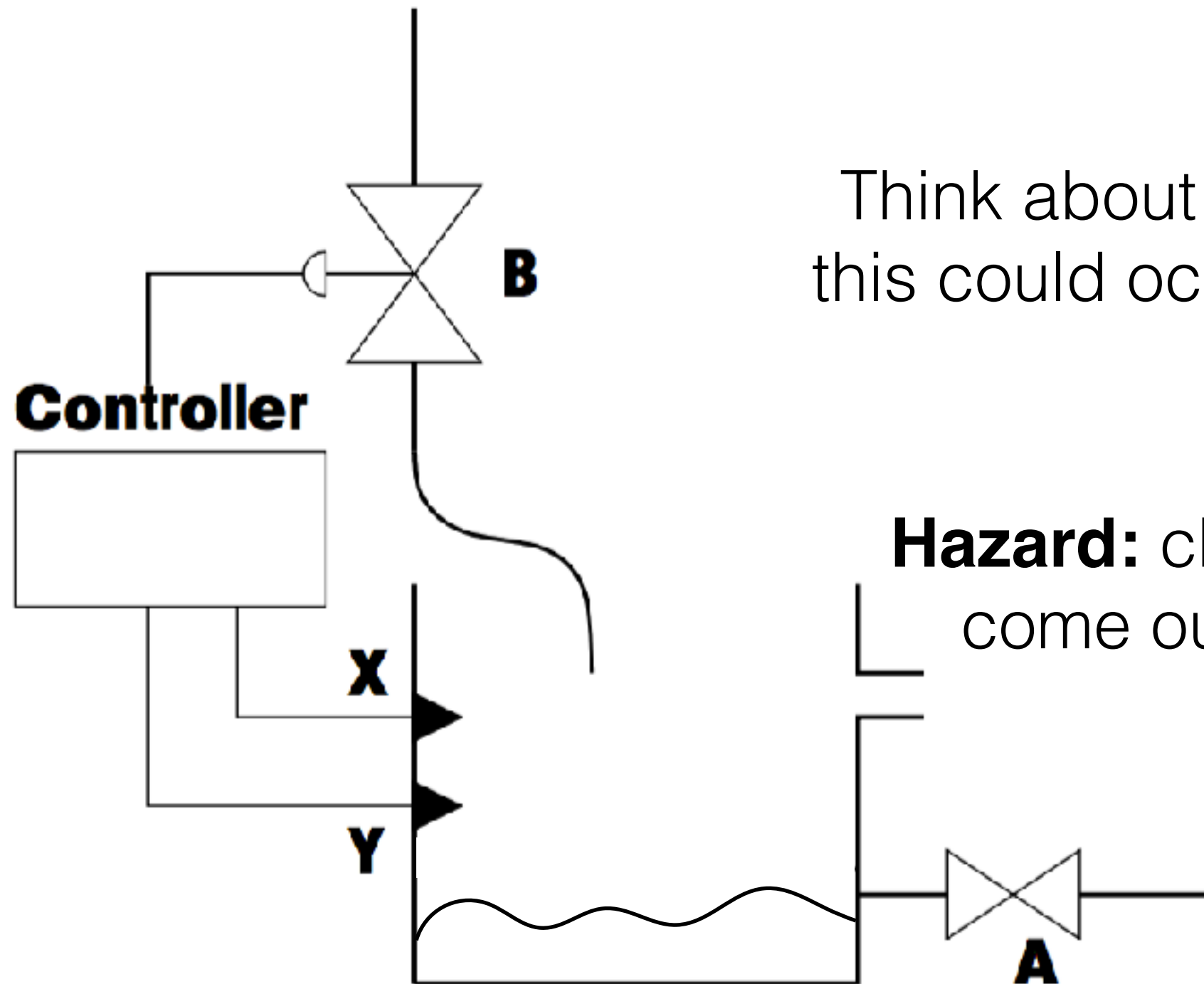
# Example: Chemical Mixing



# Example: Chemical Mixing



# Example: Chemical Mixing



Think about how  
this could occur...

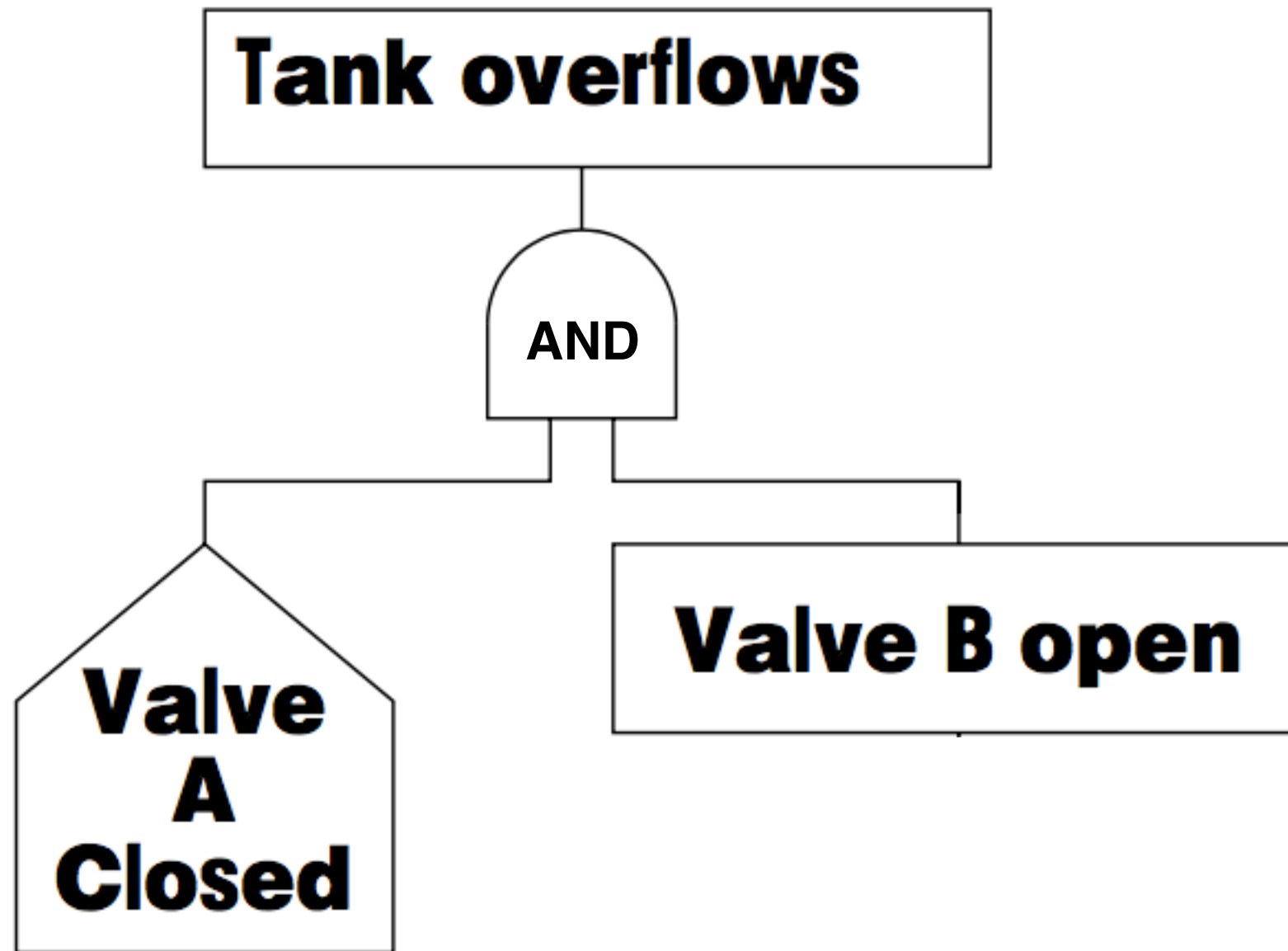
**Hazard:** chemicals  
come out here

# Fault Tree Example

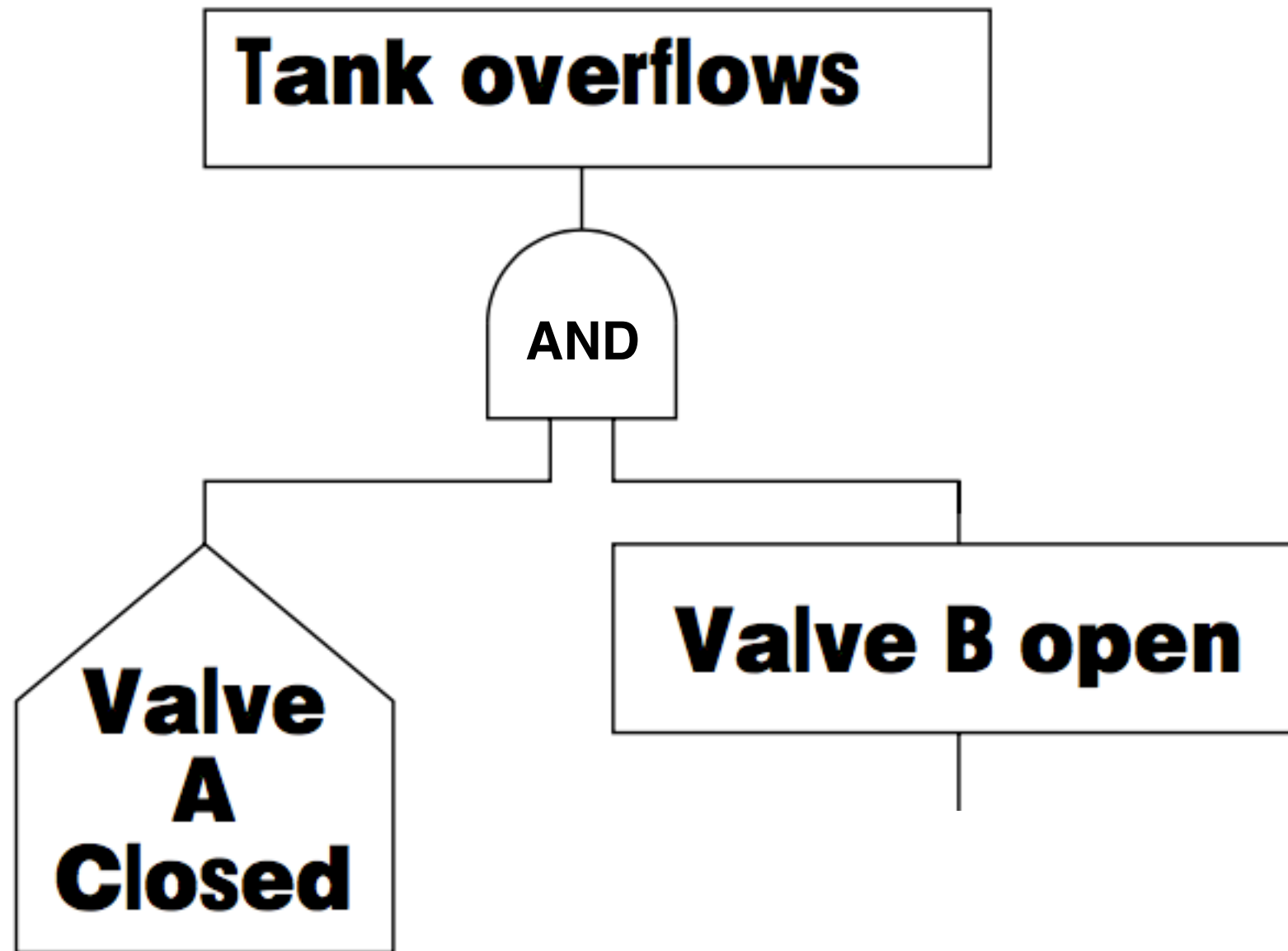
**Tank overflows**



# Fault Tree Example



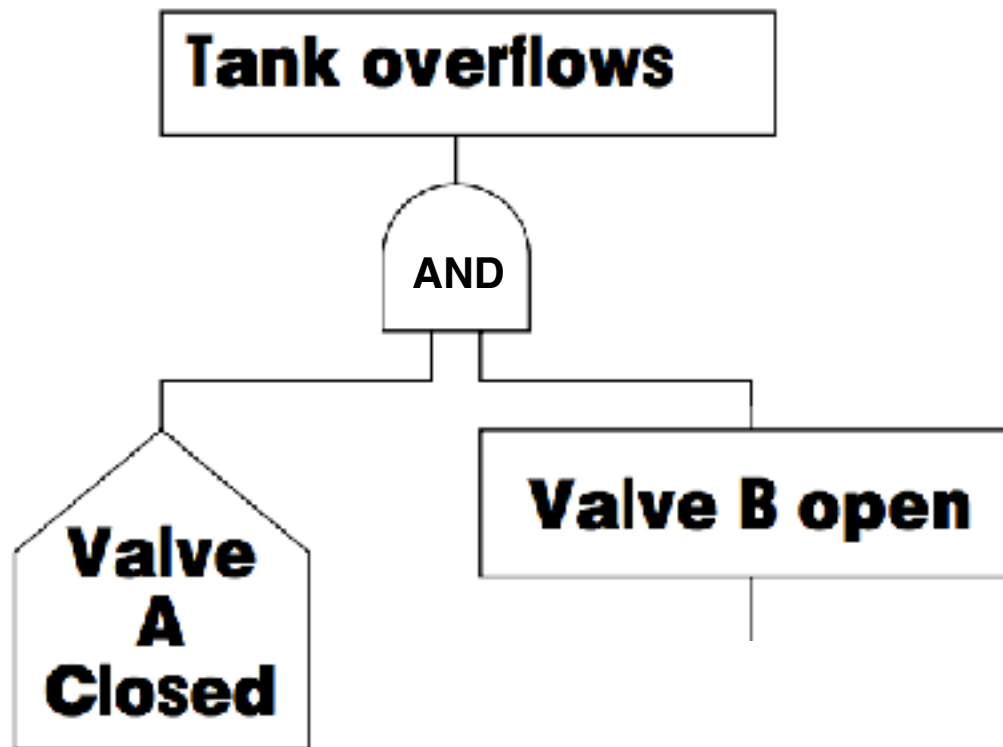
# Fault Tree Example



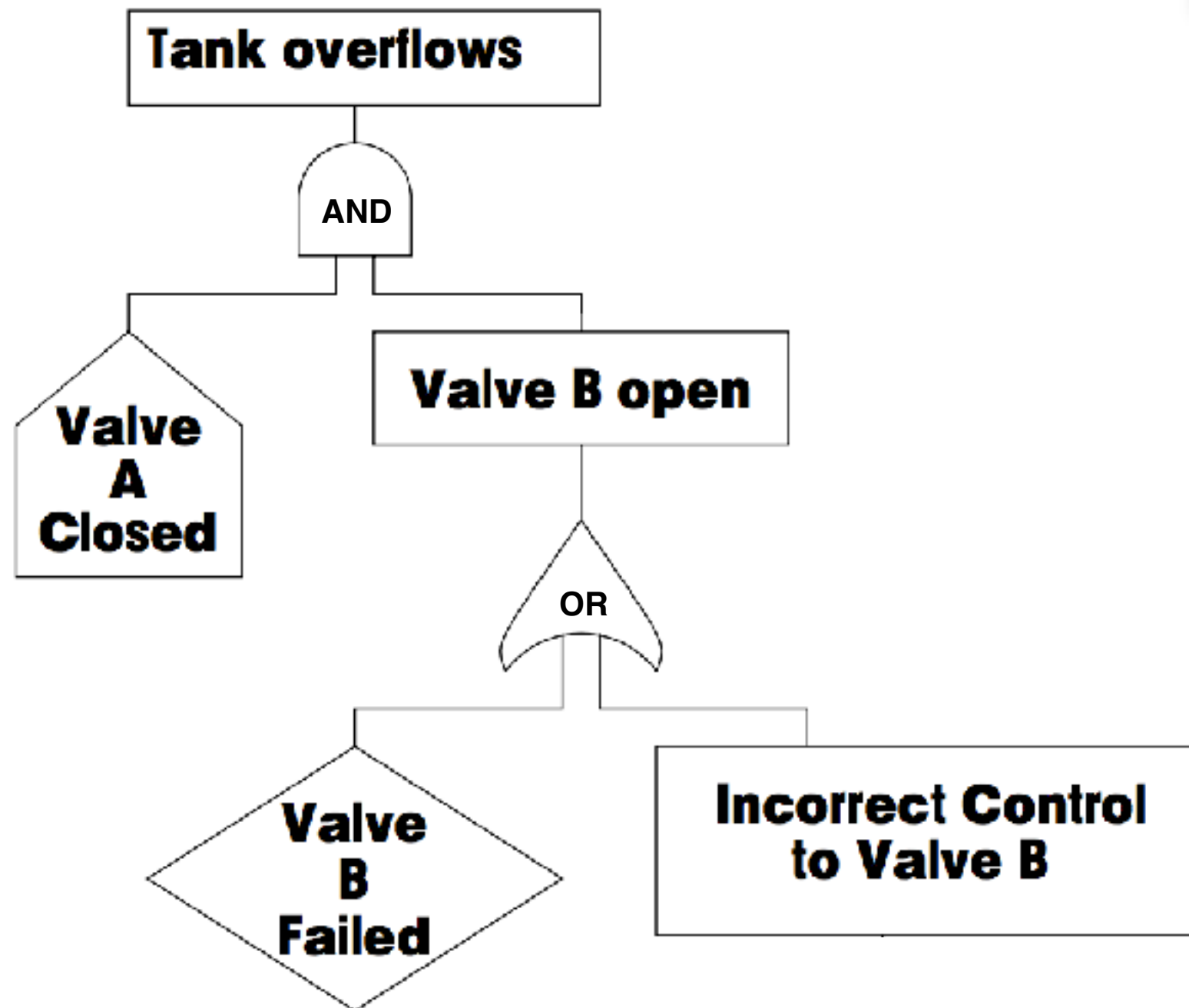
# Fault Tree Example



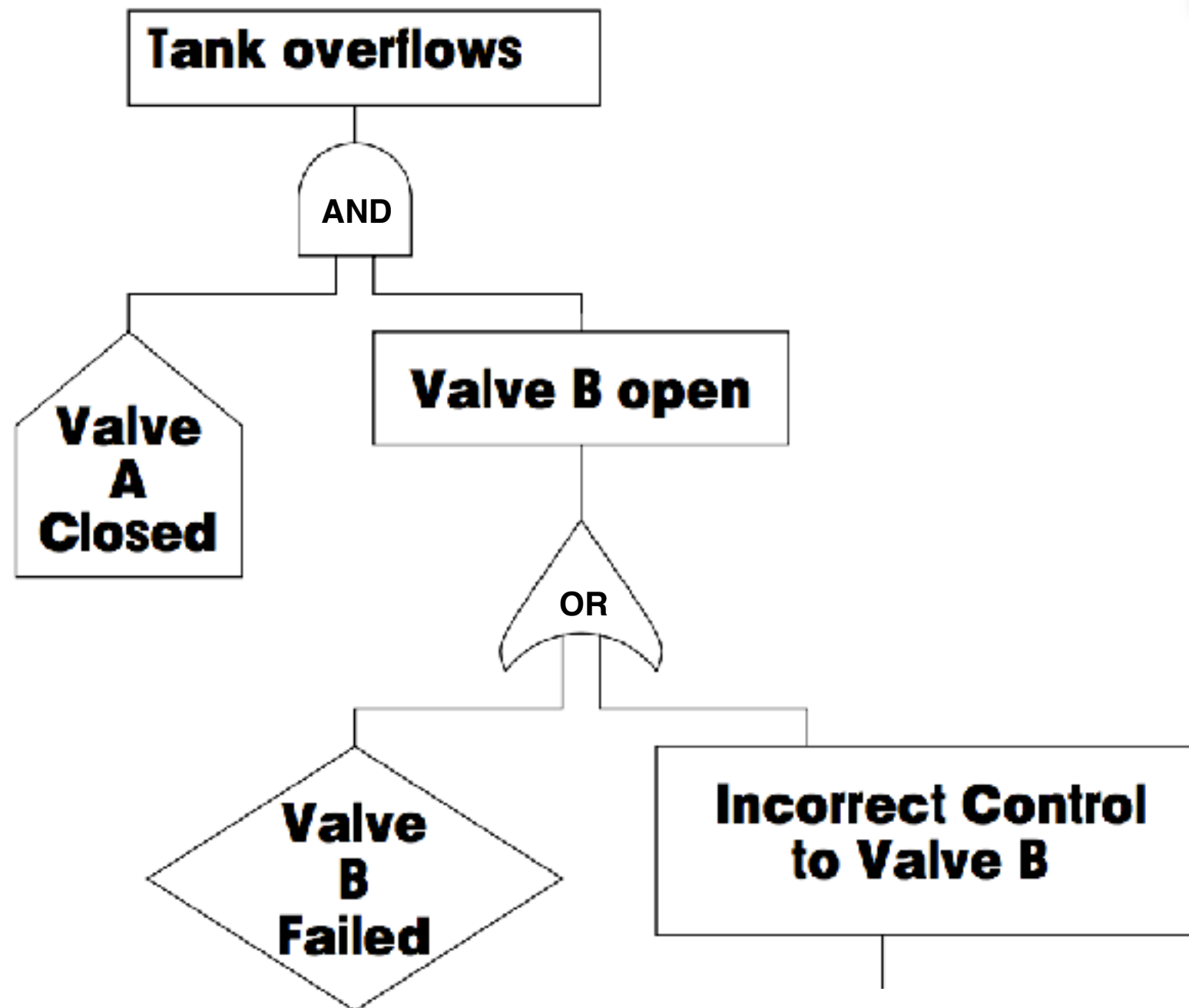
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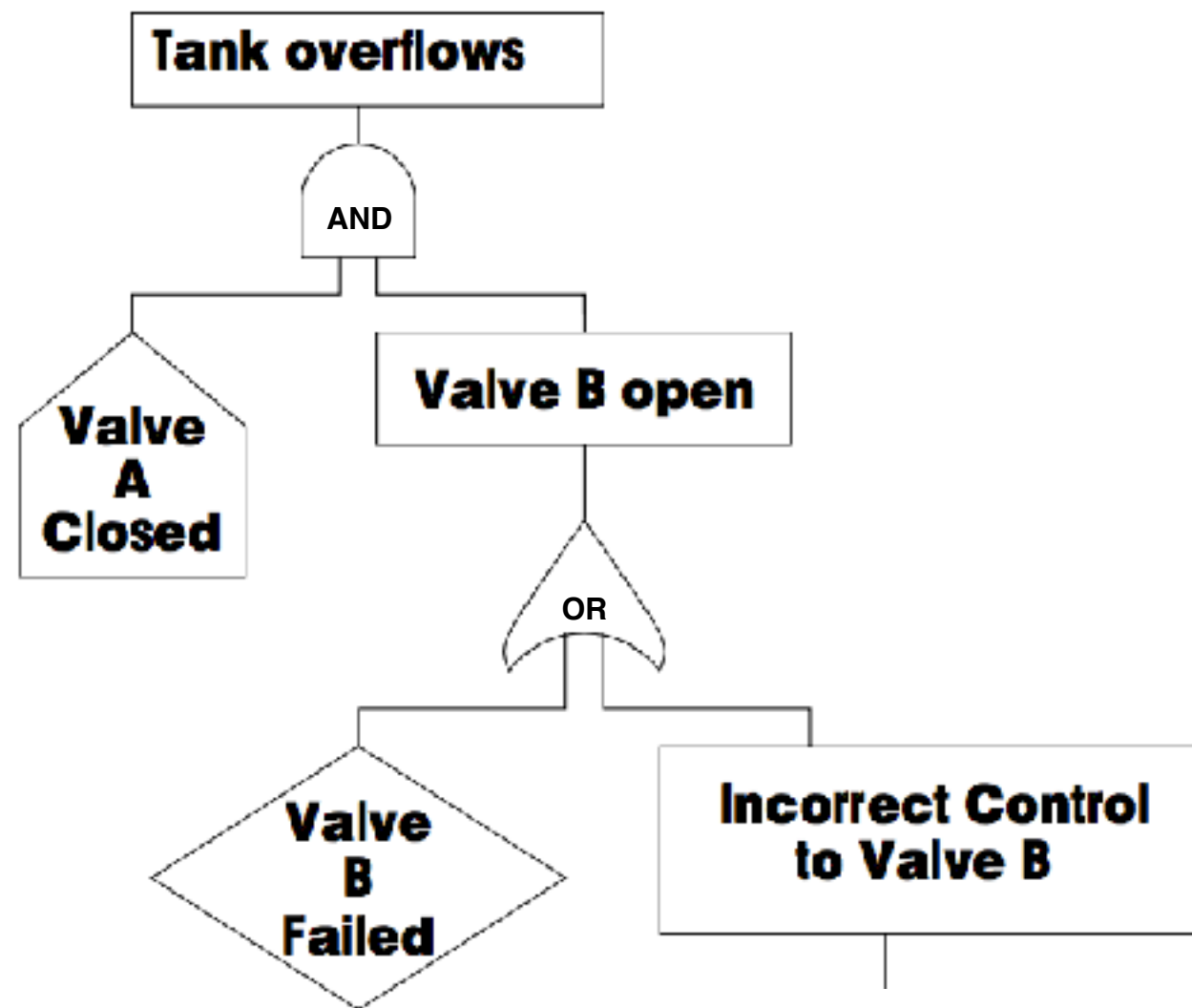
# Fault Tree Example



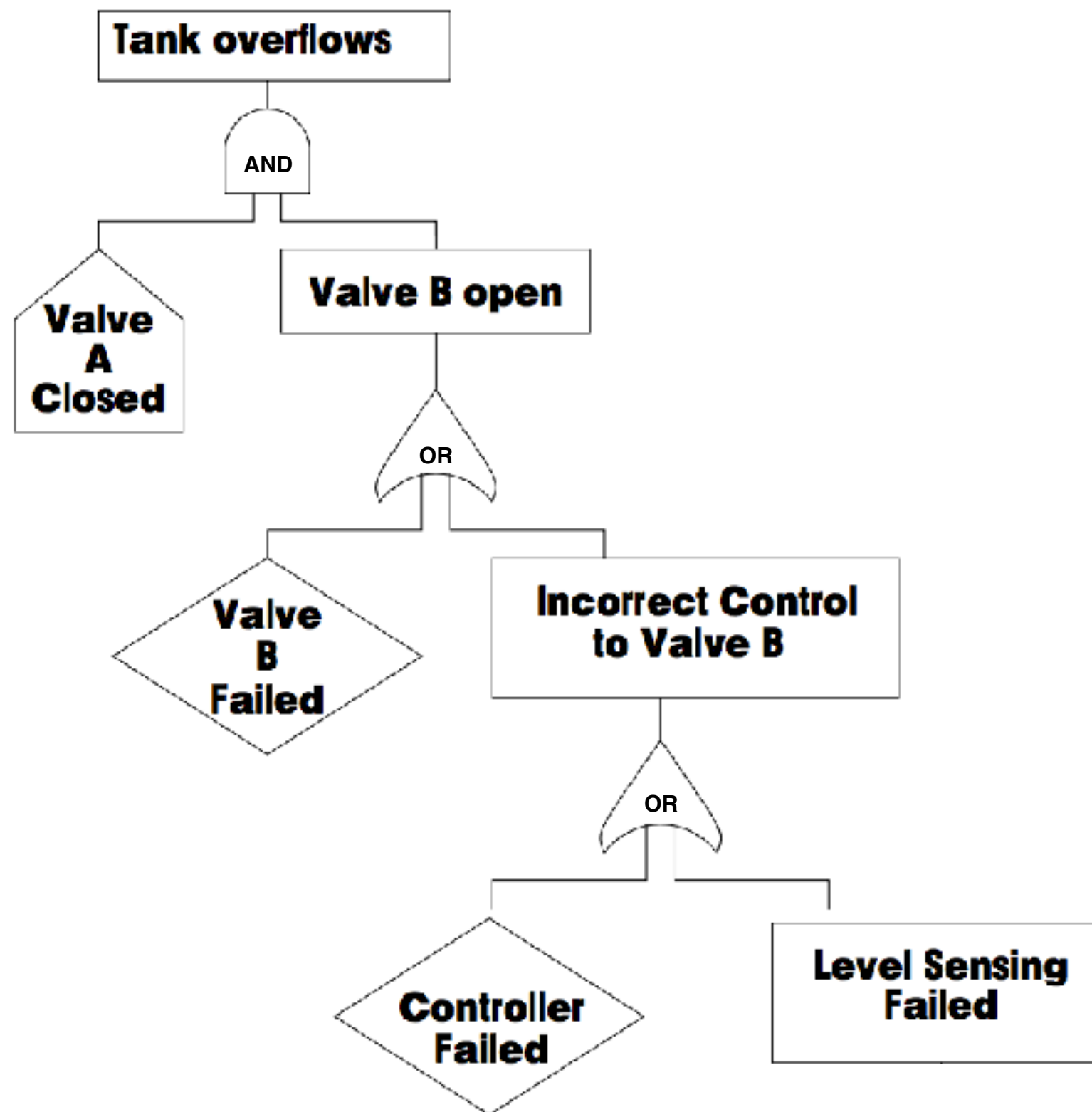
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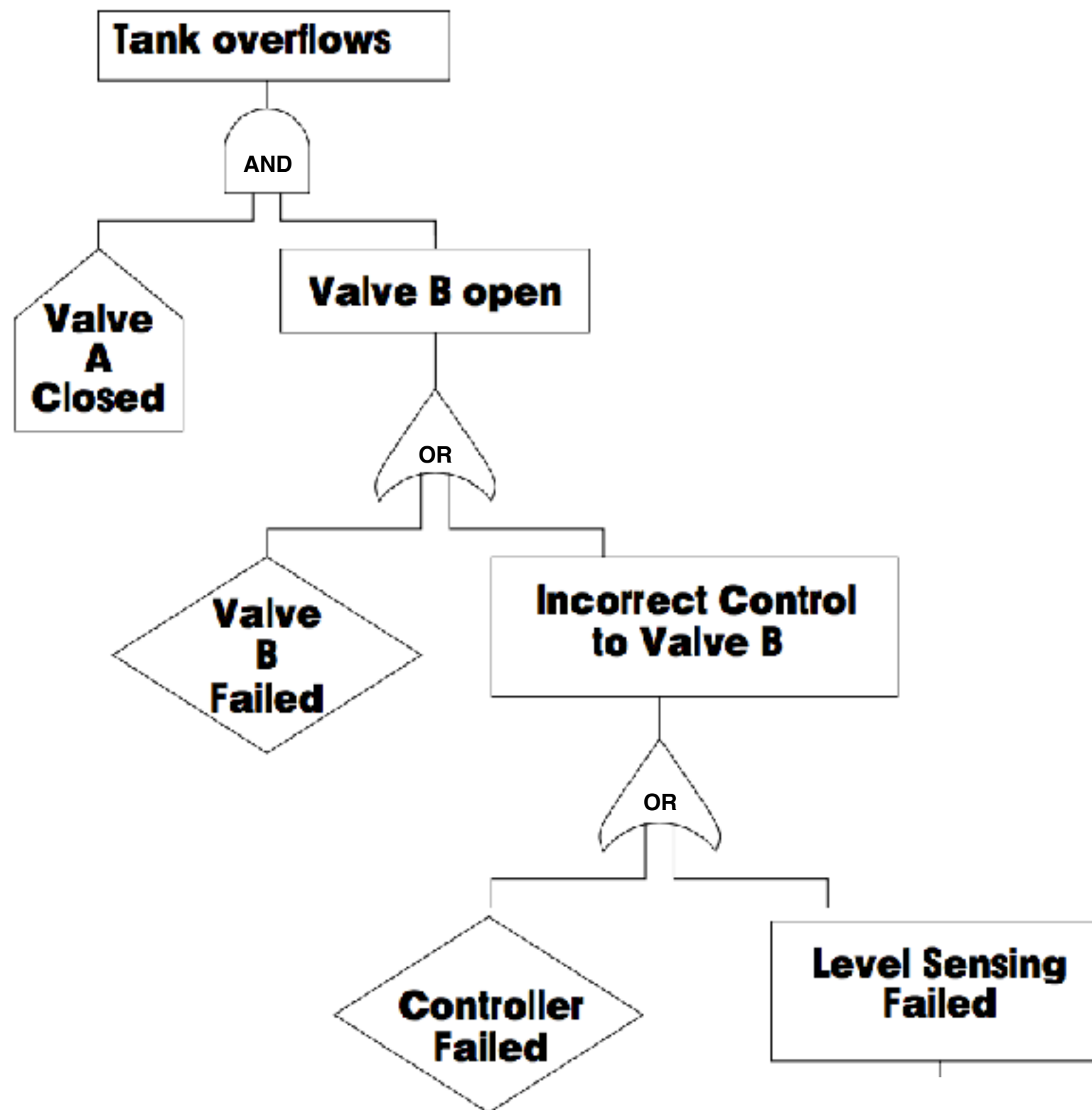
# Fault Tree Example



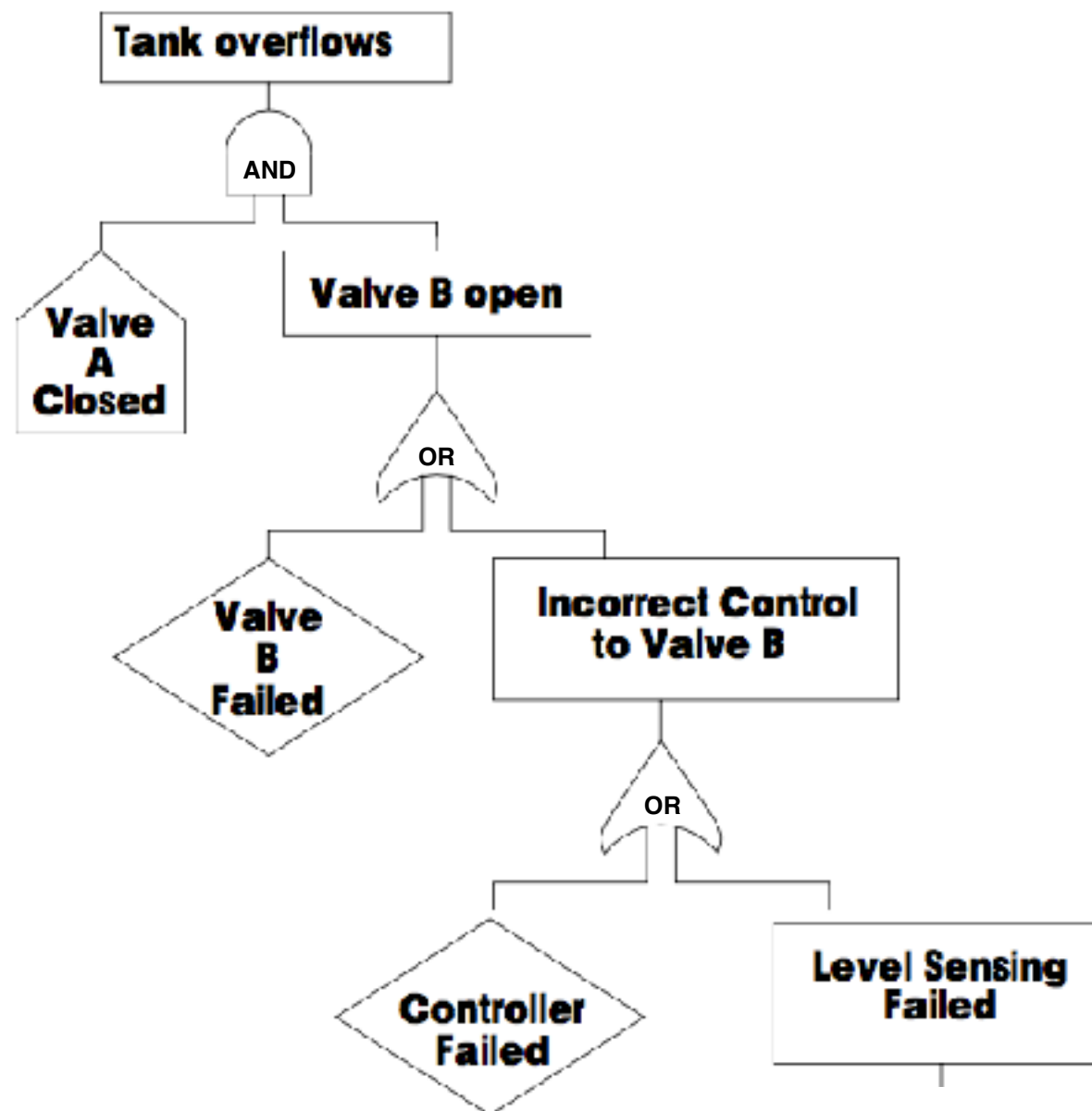
# Fault Tree Example



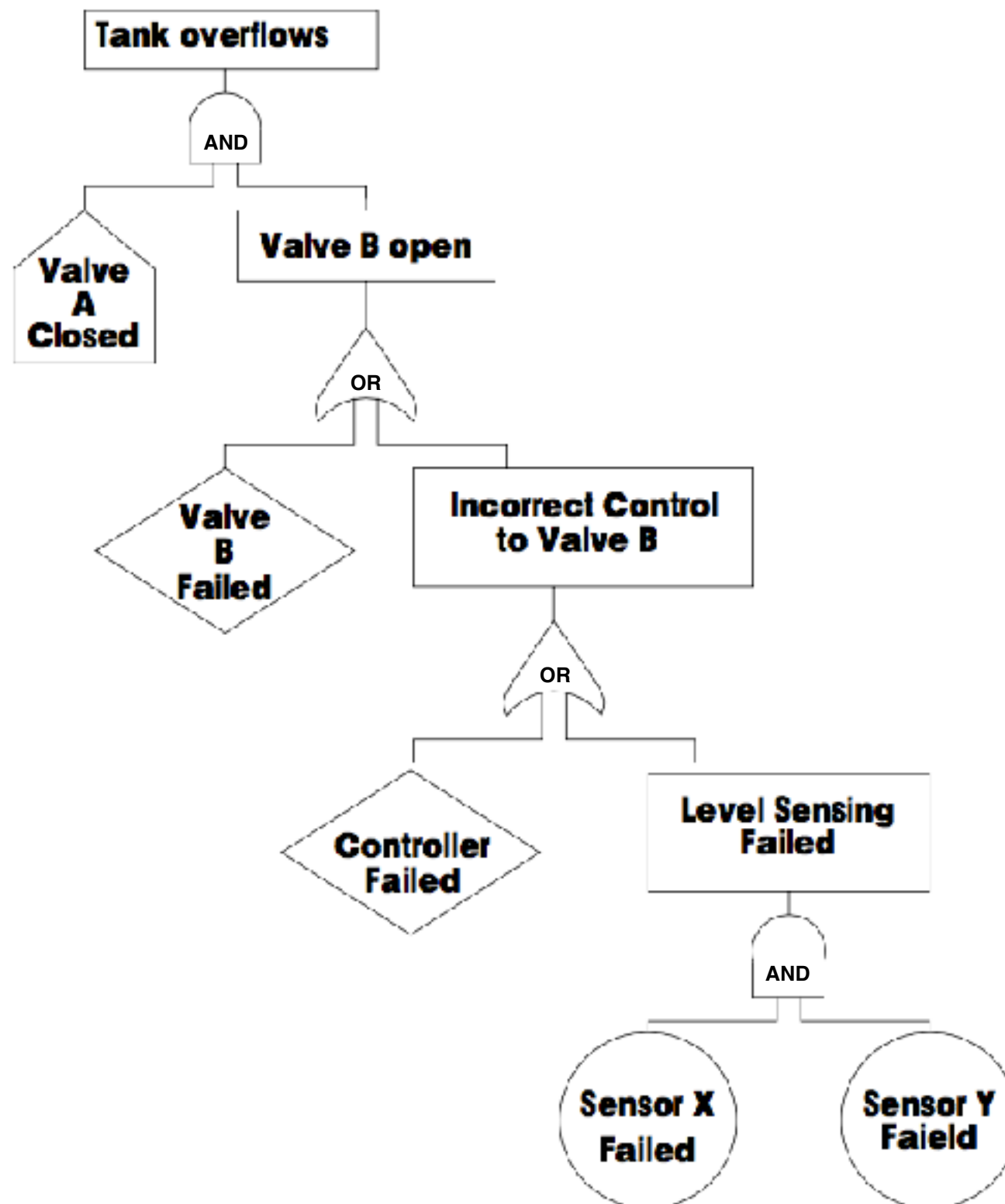
# Fault Tree Example



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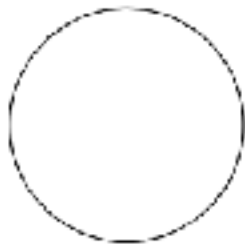


# Node Symbols



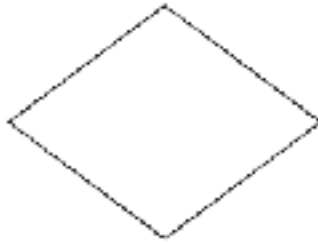
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See notes for more



## **Basic Event**

An initiating fault requiring no further action



Possibly  
analysed  
further in  
separate  
fault tree

## **Undeveloped Event**

An event which is not developed further, either because it is considered unnecessary or because insufficient information is available.



## **Intermediate Event**

An event arising from the combination of other, more basic events.



## **Normal Event**

An event which is expected to occur as part of the normal operation of the system.

# Analysis Outcomes

Causes of hazards.

## **Uses of fault-tree analysis**

to work out whether/how to change design to guard against causes.

to determine causes that the design doesn't guard against.

documentation for safety case

allow others to check our **reasoning**.

calculate probabilities of hazards arising.

See course notes for more